

Define and justify competitive fairness under lower congestion bounds setting

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Website #6 | Solver verdict: SOLVED | Verifier verdict: ISSUES | Shown: solver output

Problem

Let A be a finite set of posts, N a finite set of agents, $|A| = m$, $|N| = n$, and write

$$\ell_a := \underline{s}_a, \quad u_a := \bar{s}_a \quad (a \in A).$$

A feasible deterministic assignment is a partition $P = (S_a)_{a \in A}$ of N with congestion profile

$$s_a := |S_a|, \quad \ell_a \leq s_a \leq u_a \quad \text{for all } a, \quad \sum_{a \in A} s_a = n.$$

Agent $i \in S_a$ receives allocation (a, s_a) , and each preference $\preceq_i$ over post–congestion pairs is strictly decreasing in congestion at each fixed post.

Without lower bounds, the competitive requirement is:

$$(a, s_a) \succeq_i (x, \max\{s_x, 1\}) \quad \forall i \in S_a, \forall x \in A.$$

The task is to define the right analogue under hard lower and upper bounds (ℓ_a, u_a) , in particular to say what the benchmark at post x should be, and to justify mathematically why the definition treats forced population of posts correctly.

Alternative readings considered

I considered three readings.

- *Adopted reading (hardest natural reading).* Lower and upper bounds are hard feasibility constraints, so the fairness notion must be defined *inside* that feasible set.
- *Rejected reading: soft bounds.* One could reinterpret lower bounds as priorities or recommendations rather than hard feasibility. That is a different model, not an extension of the present one.
- *Rejected naive formula.* Replacing $\max\{s_x, 1\}$ by $\max\{s_x, \ell_x, 1\}$ does nothing, because feasibility already implies $s_x \geq \ell_x$. So that “extension” ignores lower bounds entirely.

I therefore solve the hard-bounds design problem: keep the original “take the current congestion profile as given” logic, but remove from the comparison set those outside options that are infeasible because of lower bounds.

Related literature found

- **Bogomolnaia–Moulin (2024)** — PARTIAL/BACKGROUND. This is the source of the original competitive notion in anonymous congestion: at a fixed congestion profile, agents compare to every occupied post at its current congestion and to every empty post at congestion 1. It solves the no-lower-bounds model, not the present quota extension.
- **Chen–Guo–Wen (2023)** — BACKGROUND. This paper studies the algorithmic side of the original congested-assignment competitiveness notion and notes lower/upper quotas as a nearby constrained model, but it does not provide the fairness extension needed here.
- **Ehlers–Hafalir–Yenmez–Yildirim (2014)** — TOOL/BACKGROUND. In controlled school choice with hard lower/upper bounds, they incorporate feasibility into justified envy and non-wastefulness, and they emphasize that hard bounds can destroy standard fairness/non-wastefulness while soft bounds are a distinct modeling move. I use this as conceptual guidance only.
- **Yokoi (2017)** — TOOL/BACKGROUND. In matching with lower quotas, envy-freeness is defined through feasible replacement, and lower quotas can make envy-free outcomes fail to exist. This is the closest structural analogue of the “unanimously disliked mandatory post” issue, though that paper is a two-sided matching model with acceptability constraints, not anonymous congestion.

Proof sketch

The right extension is simpler than it first appears.

The original competitive notion has two parts hidden in one formula:

1. comparison with *occupied* posts at their *current* congestion level s_x (envy-freeness style); and
2. comparison with *empty* posts at congestion 1 (non-wastefulness style).

Hard lower bounds affect these two parts differently.

For an occupied post x , nothing changes. If agent i wants the slot currently held by some agent at x , a swap preserves every post’s congestion, hence preserves all lower and upper bounds. Therefore the comparison with (x, s_x) remains the correct one.

For an empty post x , lower bounds matter. The right question is: can one populate x with one agent *while staying feasible*? In this model the answer depends only on the total slack above lower bounds,

$$\Delta := n - \sum_{a \in A} \ell_a.$$

If $\Delta = 0$, then every feasible assignment has $s_a = \ell_a$ for all a , so no agent can move to an empty post without violating some lower bound. Hence empty posts should *not* enter the competitive comparison at all. If $\Delta > 0$, then some post has one extra agent above its lower bound, and with at most one compensating reassignment any agent can be placed alone at any empty post x with $u_x \geq 1$. Therefore such an empty post should be benchmarked exactly at congestion 1.

This yields a canonical hard-bound notion: compare every agent only to *feasibly claimable* current-profile opportunities. Occupied posts are always claimable at their current congestion; empty posts are claimable at congestion 1 exactly when the system has one unit of global slack.

The “unanimously disliked mandatory post” case is then handled correctly: if agents on that post prefer some *occupied* alternative, the assignment is not competitive, because their complaint is about who bears a compulsory burden, and a feasible swap exists. If the only better alternatives are *empty* posts but $\Delta = 0$, then those comparisons are infeasible and should not count against competitiveness.

Main result

Theorem 1 (Canonical competitiveness under hard lower/upper bounds). *Let $P = (S_a)_{a \in A}$ be a feasible assignment with congestion profile s , and define*

$$\Delta := n - \sum_{a \in A} \ell_a \ (\geq 0).$$

For each post $x \in A$, define its effective competitive congestion

$$c_P(x) := \begin{cases} s_x, & s_x > 0, \\ 1, & s_x = 0, \ u_x \geq 1, \ \Delta > 0, \\ \perp, & \text{otherwise.} \end{cases}$$

A feasible assignment P is called hard-bound competitive if for every agent $i \in S_a$ and every post $x \in A$ with $c_P(x) \neq \perp$,

$$(a, s_a) \succeq_i (x, c_P(x)).$$

This notion has the following properties.

1. *It compares each agent exactly against the post-congestion pairs that are legitimate current-profile outside options under the hard bounds:*
 - *every occupied post x is compared at its current congestion s_x ;*
 - *an empty post x is compared at congestion 1 if and only if it can be activated without violating any lower/upper bound.*
2. *If all lower bounds are 0 and all upper bounds are at least 1, then*

$$c_P(x) = \max\{s_x, 1\} \quad \forall x \in A,$$

so hard-bound competitiveness reduces exactly to the original competitive assignment notion.

3. *If $\Delta = 0$, then every feasible assignment satisfies $s_a = \ell_a$ for all a , so no empty post is a legitimate comparison. In that case hard-bound competitiveness reduces to envy-freeness across occupied posts only.*
4. *If a lower bound forces some agents onto a post b and some agent $i \in S_b$ strictly prefers (x, s_x) for an occupied post x , then P is not hard-bound competitive. Thus a unanimously disliked mandatory post may make competitive assignments fail to exist; this is the correct diagnosis under hard lower bounds, not a defect of the notion.*

Proof

Lemma 1 (Occupied posts remain benchmarked at current congestion). *Let P be feasible, let $i \in S_a$, and let $x \in A$ satisfy $s_x > 0$. Then agent i can be assigned to post x while keeping the congestion of every post unchanged.*

Proof. Choose any agent $j \in S_x$. Define a new assignment Q by swapping i and j :

$$Q_x = (S_x \setminus \{j\}) \cup \{i\}, \quad Q_a = (S_a \setminus \{i\}) \cup \{j\},$$

and $Q_y = S_y$ for every $y \notin \{a, x\}$.

If $a = x$, this is just $Q = P$. If $a \neq x$, then for every post y ,

$$|Q_y| = |S_y| = s_y.$$

Hence Q has exactly the same congestion profile as P , so in particular

$$\ell_y \leq |Q_y| \leq u_y \quad \forall y \in A.$$

Thus Q is feasible, and agent i 's realized allocation in Q is (x, s_x) .

Because congestion is anonymous, the identity of j is irrelevant to i : every incumbent seat at x yields the same allocation (x, s_x) . This proves the lemma. $\square$

Lemma 2 (Characterization of activatable empty posts). *Let P be feasible, let $x \in A$ satisfy $s_x = 0$, and let $i \in S_a$. Then there exists a feasible assignment Q in which agent i is assigned to x and $|Q_x| = 1$ if and only if $u_x \geq 1$ and*

$$\Delta = n - \sum_{y \in A} \ell_y > 0.$$

Proof. We split the proof into necessity and sufficiency.

Necessity. Assume such a feasible assignment Q exists. Since $|Q_x| = 1$, feasibility of Q implies $u_x \geq 1$.

Now suppose for contradiction that $\Delta = 0$. Because P is feasible,

$$s_y - \ell_y \geq 0 \quad \forall y \in A,$$

and summing over y gives

$$\sum_{y \in A} (s_y - \ell_y) = n - \sum_{y \in A} \ell_y = \Delta = 0.$$

A sum of nonnegative integers is 0 only if each term is 0, hence

$$s_y = \ell_y \quad \forall y \in A.$$

So every post is exactly at its lower bound in P . To place agent i at the previously empty post x , some post must lose one agent overall, since the total number of agents remains n . But any post y that loses one agent would end with

$$s_y - 1 = \ell_y - 1 < \ell_y,$$

violating feasibility. This contradiction shows that $\Delta > 0$.

Sufficiency. Assume $u_x \geq 1$ and $\Delta > 0$. Since

$$\Delta = \sum_{y \in A} (s_y - \ell_y) > 0$$

and every term $s_y - \ell_y$ is a nonnegative integer, there exists at least one post $b \in A$ with

$$s_b > \ell_b.$$

We now construct a feasible assignment Q .

Case 1: $s_a > \ell_a$.

Move only agent i from a to x . Then

$$|Q_x| = 1, \quad |Q_a| = s_a - 1, \quad |Q_y| = s_y \text{ for } y \notin \{a, x\}.$$

Because $s_a - 1 \geq \ell_a$, $1 \leq u_x$, and all other post sizes are unchanged, Q is feasible.

Case 2: $s_a = \ell_a$.

Choose any agent $j \in S_b$. Since $s_b > \ell_b$, we have $b \neq a$. Move i from a to x , and move j from b to a . Then

$$|Q_x| = 1, \quad |Q_a| = s_a, \quad |Q_b| = s_b - 1, \quad |Q_y| = s_y \text{ for } y \notin \{a, b, x\}.$$

Now

$$|Q_b| = s_b - 1 \geq \ell_b, \quad |Q_a| = s_a = \ell_a, \quad |Q_x| = 1 \leq u_x.$$

All upper bounds at a and b remain satisfied because a 's size is unchanged and b 's size decreases by 1. Hence Q is feasible.

In both cases we obtain a feasible assignment Q with agent i alone at x . This proves sufficiency. $\square$

Proof of the theorem. Property (1) follows directly from the two lemmas.

For occupied posts x , Lemma 1 shows that the current-profile comparison (x, s_x) is always legitimate: a feasible swap realizes it without changing any congestion. Hence lower and upper bounds do not alter the occupied-post part of competitiveness.

For empty posts x , Lemma 2 shows that the empty-post benchmark $(x, 1)$ is legitimate exactly when the empty post can be activated feasibly. In this anonymous-congestion model, that happens exactly when $u_x \geq 1$ and the system has one unit of slack above the lower bounds, namely $\Delta > 0$. Therefore the correct competitive comparison set is exactly

$$\{(x, c_P(x)) : c_P(x) \neq \perp\}.$$

So the displayed definition of hard-bound competitiveness is precisely the feasibility-aware extension of the original competitive logic.

For property (2), if $\ell_a = 0$ for all a and $u_a \geq 1$ for all a , then

$$\Delta = n - \sum_{a \in A} \ell_a = n > 0.$$

Hence for every post x ,

$$c_P(x) = \begin{cases} s_x, & s_x > 0, \\ 1, & s_x = 0, \end{cases} = \max\{s_x, 1\}.$$

Thus the definition reduces exactly to the original one.

For property (3), if $\Delta = 0$, then the necessity part of Lemma 2 already showed that every feasible assignment must satisfy $s_a = \ell_a$ for all a . Therefore no empty post can be activated, so no empty post belongs to the competitive comparison set. The only remaining comparisons are the occupied-post comparisons (x, s_x) , i.e. envy-freeness-style comparisons across incumbents.

For property (4), suppose $i \in S_b$ and some occupied post x satisfies

$$(x, s_x) \succ_i (b, s_b).$$

By Lemma 1, (x, s_x) is a legitimate comparison for agent i . Therefore the defining inequality of hard-bound competitiveness fails for i , so P is not hard-bound competitive.

This last point is exactly how the definition should behave. Hard lower bounds do *not* erase justified envy toward occupied attractive posts, because a feasible swap preserves all bounds. What lower bounds do erase are only those “outside options” that are infeasible solely because opening an empty post would force some other post below its floor. $\square$

Corollary 1 (Interpretation of a forced unpopular post). *Let P be feasible and let $b \in A$ satisfy $\ell_b > 0$.*

1. *If an agent $i \in S_b$ strictly prefers some occupied post benchmark (x, s_x) , then P is not hard-bound competitive.*
2. *If all posts that i strictly prefers to (b, s_b) are empty and $\Delta = 0$, then those preferences do not by themselves refute hard-bound competitiveness, because those empty posts are infeasible outside options.*

Proof. Part (1) is property (4) of the theorem.

For part (2), if $\Delta = 0$, then by the theorem every empty post has $c_P(x) = \perp$, so empty posts are excluded from the comparison set. Thus only occupied posts can witness non-competitiveness. $\square$

Gap to the full statement

None. The theorem gives a complete hard-bound extension of competitiveness, specifies exactly what replaces $(x, \max\{s_x, 1\})$, proves reduction to the original notion, and explains how forced population of unpopular posts is treated.

Self red-team

- **Counterexample attempt 1:** one mandatory bad post b , one attractive empty post x , and $\Delta = 0$. I checked the smallest instance $n = 1$, $\ell_b = 1$, $\ell_x = 0$. My definition makes x non-comparable, so the unique feasible assignment can still be competitive. This is correct: the agent cannot move to x without violating the floor at b .
- **Counterexample attempt 2:** same picture, but $\Delta > 0$. Then the empty attractive post becomes comparable at congestion 1. The definition declares the assignment non-competitive if the agent prefers $(x, 1)$. This is correct: the empty post is now feasibly activatable.
- **Counterexample attempt 3:** a mandatory bad post b and an occupied good post g . The definition keeps (g, s_g) as a benchmark, so agents forced to b may still have justified envy. This is intended: lower bounds do not decide *which* agents must bear the burden, and a feasible swap exists.

- **Plausible objection 1:** Why not compare to $(x, s_x + 1)$ for occupied posts, since an actual unilateral move would increase congestion? Answer: that would be Nash stability, not competitiveness. The original competitive notion is envy/non-wastefulness at a fixed congestion profile, and a swap with an incumbent preserves the current congestion at x .
- **Plausible objection 2:** Why may agent i 's empty-post complaint rely on slack from another post $b \neq a$? Answer: because lower bounds constrain only post populations, not which particular agents occupy them. In this one-sided model every agent can serve as the compensating reassignment, so the correct question is global feasibility, not local slack at i 's current post.
- **Plausible objection 3:** Does the notion hide the unfairness of a unanimously disliked mandatory post? Answer: no. If there is an occupied better seat elsewhere, the notion declares non-competitiveness. It ignores only *infeasible* empty-post fantasies when all floors already bind.
- **Boundary cases checked.**
 - If all $\ell_a = 0$, the formula reduces to $\max\{s_x, 1\}$.
 - If $u_x = 0$, then x is never a legitimate benchmark at congestion 1.
 - If $m = 1$, there are no cross-post comparisons; the definition is tautologically satisfied by the unique feasible assignment.
 - If $\Delta = 0$, every feasible assignment has $s = \ell$, as used in the proof.

Invoked results

No external theorem is used as a proof step. All cited papers are used only for background, orientation, and conceptual comparison, so there is no hypothesis-match audit to record inside the proof itself.

Self-confidence

3 — The mathematical core is elementary and I checked every step, but the word “canonical” still rests on the modeling judgment that competitiveness should compare only to feasible current-profile outside options.

Bibliography

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\begin{verbatim}
@misc{BogomolnaiaMoulin2024,
  author      = {Anna Bogomolnaia and Herve Moulin},
  title       = {Fair Congested Assignment Problem},
  year        = {2024},
  eprint      = {2301.12163},
  archivePrefix= {arXiv},
  primaryClass= {econ.TH},
  doi         = {10.48550/arXiv.2301.12163}
}

@misc{ChenGuoWen2023,
```

```

author      = {Jiehua Chen and Jiong Guo and Yinghui Wen},
title       = {Algorithms and Complexity for Congested Assignments},
year        = {2023},
eprint      = {2312.07431},
archivePrefix= {arXiv},
primaryClass = {cs.GT},
doi         = {10.48550/arXiv.2312.07431}
}

@article{EhlersHafalirYenmezYildirim2014,
author      = {Lars Ehlers and Isa E. Hafalir and M. Bumin Yenmez and
              Muhammed A. Yildirim},
title       = {School Choice with Controlled Choice Constraints:
              Hard Bounds versus Soft Bounds},
journal      = {Journal of Economic Theory},
volume      = {153},
pages       = {648--683},
year        = {2014},
doi         = {10.1016/j.jet.2014.03.004}
}

@inproceedings{Yokoi2017,
author      = {Yu Yokoi},
title       = {Envy-free Matchings with Lower Quotas},
booktitle    = {28th International Symposium on Algorithms and Computation
              (ISAAC 2017)},
series      = {Leibniz International Proceedings in Informatics (LIPIcs)},
volume      = {92},
pages       = {67:1--67:12},
year        = {2017},
doi         = {10.4230/LIPIcs.ISAAC.2017.67}
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